Dynamic-Capacity Relational Graph Convolutional Networks for Large-Scale Biomedical Knowledge Graph Completion

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***Abstract*—**

Clinical knowledge graph completion is an important step toward improving the consistency, completeness, and validity of large-scale medical ontologies. This paper proposes a Dynamic-Capacity Relational Graph Convolutional Network (R-GCN) for scalable biomedical knowledge graph learning. In this framework, a composite clinical graph is constructed by integrating curated semantic relationships from SNOMED CT with empirically derived diagnostic co-occurrence relations from MIMIC-IV electronic health records. To effectively handle the scale and heterogeneity of the integrated graph, our model employs degree-aware (dynamic-informed) dynamic-capacity node embeddings together with node-centric subgraph sampling for memory-efficient training. This structure-centric design enables scalable relational learning without relying on pretrained language models, while addressing embedding heterogeneity and graph scalability challenges in large biomedical knowledge graphs. On integrated SNOMED-MIMIC graphs, the proposed Pure R-GCN substantially outperforms embedding-based and text-enhanced baselines, achieving over 3 times improvement in Mean Reciprocal Rank over ComplEx and over 2 times improvement over a Hybrid RoBERTa + R-GCN model, demonstrating significant gains in multi-relational link prediction performance.

Keywords*— *Biomedical knowledge graphs, SNOMED CT, MIMIC-IV, Relational Graph Convolutional Networks, Link prediction, Dynamic-capacity embeddings, Subgraph sampling.

I. INTRODUCTION

Knowledge graphs provide a powerful framework for representing complex relationships between entities. Many impactful healthcare applications rely on knowledge graphs to organize and reason over structured medical knowledge. In particular, medical ontologies serve as structured vocabularies that standardize and precisely express clinical concepts and their interrelationships, enabling interoperability, clinical decision-making, and large-scale analytics. These ontologies form semantic backbones of electronic health records (EHRs), supporting cohort definitions, population-level analyses, and rule-based as well as learning-based clinical decision support systems.

Among all ontologies, the Systematized Nomenclature of Medicine - Clinical Terms (SNOMED CT) is considered to have the most comprehensive and widely adopted clinical terminology worldwide [1] . It comprises over 350,000 medical concepts connected by more than one million semantic relationships, with consistent representation of clinical knowledge across electronic health records, decision-support systems, and biomedical research platforms. As a result, SNOMED CT plays a central role in clinical data interoperability, quality measurement, and secondary use of EHR data for analytics and research.

Despite its broad utilities, SNOMED CT remains incomplete and occasionally inconsistent, particularly in its relational structure. Many medical notions lack explicit, correctly defined relationships, while certain hierarchical links, especially is-a relationships, are either missing or misclassified. An analysis of an extracted SNOMED CT snapshot shows that the is-a relationship accounts for approximately 47.3% of all active relations, taking a dominant role of hierarchical structure in the ontology [3]. These omissions weaken the ontology's semantic integrity, leading to potential issues such as incomplete cohort extraction, incorrect aggregation of clinical concepts, and reduced reliability of EHR-based analytics. Beyond hierarchical relations, other semantic links, including morphological, temporal, and structural relations, are also underrepresented, further limiting the ontology's expressiveness [3], [4].

Incomplete ontologies in healthcare informatics compromise both clinical decision support (CDS) and the results of EHR-

driven analytics. Many CDS systems rely on SNOMED CT's relational structure to infer clinical context, aggregate related diagnoses, and trigger alerts or recommendations. Missing or weakly defined relationships can result in false negatives, incomplete rule activation, or inconsistent patient stratification. Similarly, cohort definition and population health studies depend on accurate semantic relationships to pool related conditions and procedures [3]. As a result, deficiencies in ontology structure propagate from terminological gaps into clinical workflows, analytics pipelines, and quality-of-care assessments [4].

These limitations arise from two primary factors: dataset-level constraints and algorithmic bottlenecks. At a data level, biomedical ontologies such as SNOMED CT lack sufficient empirical coverage when used in isolation, failure to capture non-hierarchical and clinically contextual relationships that arise in real-world practice. Methodologically, existing graph-based learning models are hindered by fixed-dimensional embeddings and full-graph processing. These architectural choices limit their capacity to represent heterogeneous relation types and scale efficiently to knowledge graphs containing hundreds of thousands of medical concepts. To address these challenges, we employ large-scale multi-relational link prediction to identify missing or weakly defined relationships and support ontology enrichment for downstream clinical analytics. This study integrates SNOMED CT with the MIMIC-IV (Medical Information Mart for Intensive Care IV) dataset, a large-scale electronic health-record (EHR) corpus capturing real-world diagnostic, laboratory, and clinical observations [2]. While SNOMED CT provides a curated and standardized semantic framework, MIMIC-IV provides the empirical, contextual, and temporal evidence derived from routine clinical care. From an informatics perspective, this integration supports clinical knowledge maintenance and validation, enables empirical EHR evidence to inform and refine ontology structure in ways that directly benefit cohort construction, clinical workflow analytics, and decision-support systems.

Learning over biomedical graphs of this magnitude is non-trivial. Traditional graph neural networks (GNNs), including Relational Graph Convolutional Networks (R-GCNs), assume uniform embedding sizes for all nodes and operate on the full adjacency graph. This approach is computationally infeasible for ontologies with highly skewed degree distributions. In SNOMED CT, a small subset of high-degree concepts (e.g., Disease, Clinical finding) dominates the connectivity structure, while the majority of concepts occur infrequently. As a result, uniform embedding strategies overlook this inherent heterogeneity, disproportionally exhausting representational capacity on low-impact nodes. On the other hand, full-graph message passing leads to the out-of-memory (OOM) problem when applied to graphs containing millions of edges. The SNOMED-MIMIC graph contains hundreds of thousands of nodes and millions of edges, making full-graph training infeasible on commodity GPUs.

In this paper, we propose a Dynamic-Capacity Relational Graph Convolutional Network (R-GCN) that integrates relational information derived from ontology and EHR. This network allocates embedding capacity based on node degree and enables scalable training through memory-safe node-centric subgraph sampling. The main contributions of this work are summarized as follows:

1) Dynamic-Capacity node embeddings: We introduce a degree-aware embedding strategy that assigns larger embedding dimensions to high-degree clinical concepts and compact embeddings to low-degree nodes, projecting all embeddings into a shared latent space to balance expressiveness and memory efficiency.
2) Memory-saving in node-centric subgraph sampling: We develop a subgraph extraction strategy that expands local neighborhoods around sampled edges while enforcing strict caps on node and edge counts, enabling scalable R-GCN training without loading the full SNOMED CT + MIMIC-IV graph into GPU memory.
3) Unified multi-relational clinical graph construction: We construct a heterogeneous clinical knowledge graph that combines curated SNOMED CT semantic relations with empirical diagnostic co-occurrence relations derived from ICD-10 diagnoses mapped to SNOMED CT via OHDSI Athena.

Our proposed model predicts missing or weakly defined relationships across all SNOMED CT relation types, not limited to hierarchical links, and supports ontology refinement using empirical ICU patient data. Unlike text-centric approaches, our method directly exploits the topological and multi-relational properties of biomedical knowledge graphs, enabling more expressive and semantically grounded relational learning for healthcare informatics applications.

II. Related work

Recent research has explored deep-learning approaches to enrich and validate SNOMED CT. Prior methods treated relation detection as a binary *is-a* versus *non-is-a* task using embedding bags for variable-length inputs and transformer encoders for contextual semantics, achieving F1 scores of up to 93% [6], [8]. Multi-relation classification methods extended this setting to 133 relation types using TransE-based knowledge graph embeddings combined with a multi-head classifier, reporting approximately 91% test accuracy [7]. Together, these approaches demonstrated meaningful progress but revealed two key limitations. First, fixed embedding dimensionality may be insufficient to capture complex hierarchical semantics. Second, scalability remains largely untested on large, cross-domain clinical graphs.

Earlier work on link prediction similarly relied on single, static embedding spaces, which restrict relational expressiveness in large graphs [12]. Likewise, scalable feature learning methods continue to depend on fixed-dimension embeddings [20]. More

recent studies reaffirm these limitations in biomedical settings. Knowledge graph embedding models for ICU readmission prediction showed that increasing embedding dimensionality beyond 300 yields no performance gain while substantially increasing computational cost [9]. Similarly, automated SNOMED CT concept annotation models using Bi-GRU architectures and large contextual embeddings reported reduced scalability when uniformly representing all concepts.

III. DATASETS

This study uses SNOMED CT (U.S. Edition, September 1, 2025, Snapshot),which is a comprehensive clinical terminology developed and maintained by the U.S. National Library of Medicine (NLM), in coordination with SNOMED International, which governs the international standard. The SNOMED CT U.S. Edition is released on a biannual update cycle (March and September), integrating the corresponding SNOMED CT International Release with U.S.-specific extensions. We use the September 2025 U.S. Edition snapshot (effectiveTime = 20250901), as it was the latest stable release available at the time of this study. The terminology comprises over 350,000 medical concepts and millions of formally defined semantic relationships, including both hierarchical and non-hierarchical relations [1].

We also use MIMIC-IV (version 3.1), a large-scale, de-identified critical care database maintained by the MIT Laboratory for Computational Physiology and distributed via PhysioNet under the credentialed health data license. The dataset contains detailed electronic health records from intensive care unit admissions, including diagnoses, procedures, laboratory measurements and clinical outcomes. MIMIC-IV is released in versioned updates as data quality improvements and schema refinements are introduced. Version 3.1 was selected because it represents the most recent publicly available release at the time of analysis and is consistent with current clinical coding standards [2].

Both datasets serve as the source material for constructing the clinical knowledge graph used in this work.

A. SNOMED CT Relation and Description Extraction

The SNOMED CT release provides multiple structured components that support semantic graph construction. In this study, we primarily utilize the Relationship and Description tables from the SNOMED CT Snapshot distribution.

The Relationship table (sct2_Relationship_Snapshot) encodes directed semantic relationships between clinical concepts. Each record specifies a source concept identifier, a destination concept identifier, and a relationship type identifier, along with associated metadata such as activity status and relationship group. These relationships capture both hierarchical and non-hierarchical semantics essential for clinical reasoning. The Description table (sct2_Description_Snapshot) provides textual representations for each concept, including fully specified names, preferred terms, and synonymous labels. This table enables consistent concept labeling and supports interpretability during graph analysis.

Prior to graph construction, the SNOMED CT data were cleaned and standardized using a dedicated preprocessing pipeline. Only active relationships were retained to ensure clinical relevance and consistency. For each concept, preferred terms were selected using the designated description type identifier (typeId = 900000000000013009). Relationship type identifiers were mapped to human-readable semantic labels (e.g., 116680003 → is-a, 47429007 → associated-with) to facilitate downstream analysis. Records containing missing, incomplete, or corrupted fields were removed. Summary statistics were logged before and after preprocessing to ensure reproducibility and traceability of the data preparation process.

Table I: Examples of SNOMED CT relations extracted after preprocessing.

source_id	source_term	relation_term	type_id	destination_id	destination_term
400932004	Binocular perimetry	Procedure site - Direct	405813007	49549006	Visual system structure
295858004	Intentional dothiepin overdose (disorder)	Causative agent (attribute)	246075003	373873005	Pharmaceutical / biologic product
95208000	Photic epilepsy	is-a (attribute)	116680003	29753000	Local convulsion
17859000	Polyvalent crotalid ae antivenin (substance)	is-a (attribute)	116680003	71289008	Product containing snake antivenom immunoglobulin

The resulting output consists of a cleaned and semantically normalized set of SNOMED CT concepts and relations, which serves as the ontology backbone for subsequent clinical knowledge graph construction.

This extraction step provides the structured semantic relationships required for clinical knowledge graph construction, ensures interpretable and human-readable relation labels for downstream analysis, and establishes the ontology backbone onto which empirical relations derived from MIMIC-IV can be integrated.

MIMIC-IV ICD to SNOMED CT Mapping via OHDSI Athena:

Diagnosis information was extracted from the hosp/diagnoses_icd.csv.gz table in MIMIC-IV. To align with the coverage of the selected SNOMED CT release and current clinical coding standards, only ICD-10-CM diagnoses were retained, while ICD-9-CM codes were excluded.

To map ICD-10-CM diagnoses to SNOMED CT concepts, we employed the OHDSI Athena vocabulary repository, which provides curated cross-terminology mappings maintained by clinical domain experts. Athena was selected because it offers standardized and version-synchronized mappings compatible with both SNOMED CT and UMLS vocabularies.

The mapping process utilized two Athena tables: CONCEPT, which defines concept identifiers, names, vocabularies, and domains, and CONCEPT_RELATIONSHIP, which encodes inter-vocabulary mappings. Relationships labeled “Maps to” and “Maps to value” were extracted and filtered to retain mappings where the source vocabulary was ICD-10-CM and the target vocabulary was SNOMED CT. These mappings were joined with textual concept names for interpretability and merged with MIMIC-IV diagnosis records.

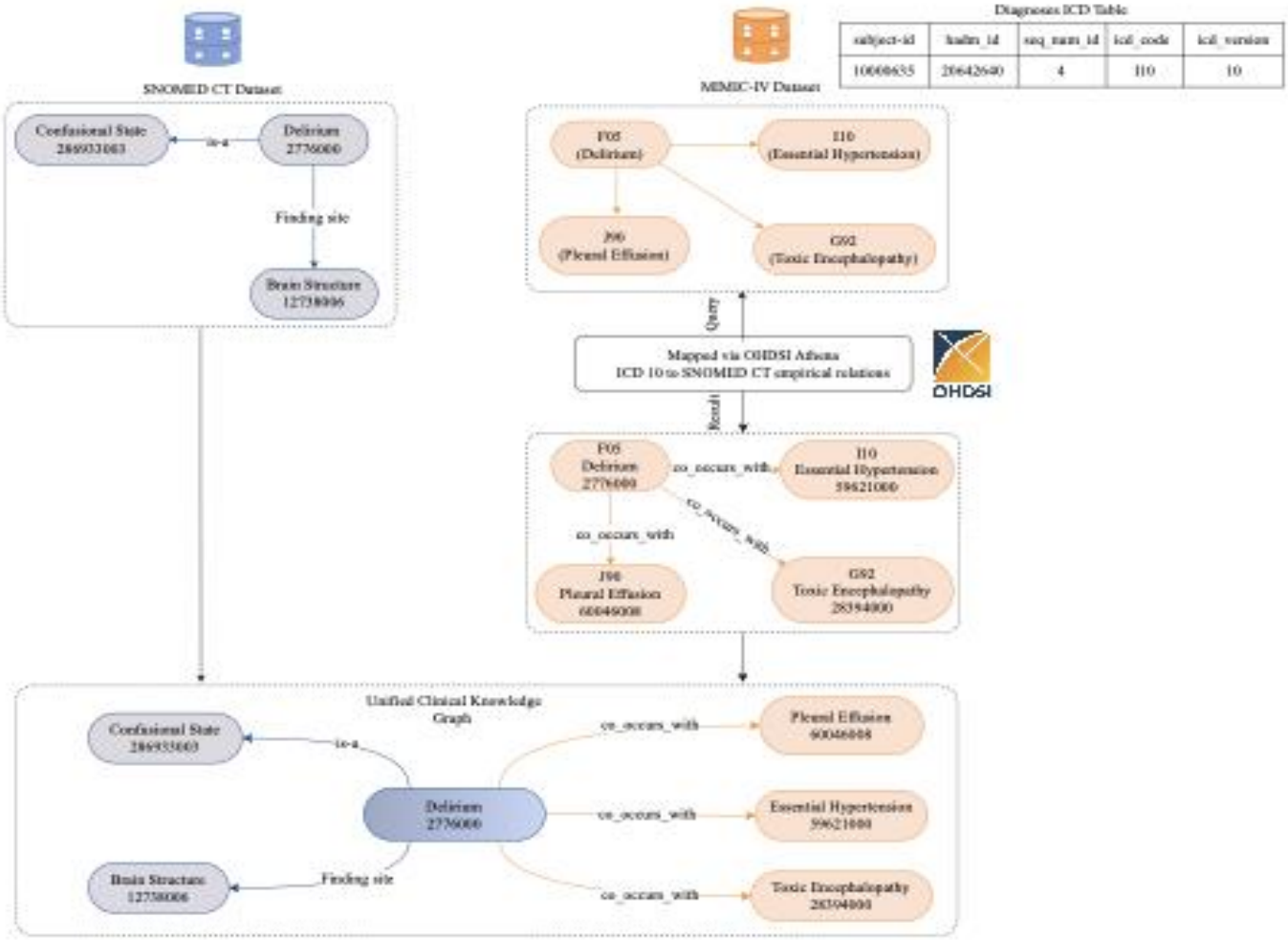

Figure 1. SNOMED CT and MIMIC-IV Integration via OHDSI Athena

This procedure produces SNOMED CT, aligned diagnostic associations that can be directly integrated into the ontology graph, enabling empirical clinical data from MIMIC-IV to participate in SNOMED CT's multi-relational structure.

Table II: ICD-10 Diagnoses Mapped to SNOMED CT Concepts via OHDSI Athena

sourceId	source_term	relation_term	destination_id	destination_term
18726006	Senile asthenia	co_occurs_with	195029002	Cardiomyopathy associated with another disorder
127785005	Active or passive immunization	co_occurs_with	31097004	Post poliomyelitis syndrome
360549009	Severe protein-calorie malnutrition	co_occurs_with	49049000	Parkinson's disease
127785005	Active or passive immunization	co_occurs_with	719848005	Disorder caused by alcohol

B. SNOMED CT and MIMIC-IV Multi-Relational Graph Construction

After standardizing both SNOMED CT and MIMIC-IV diagnoses to unified SNOMED CT concept identifiers, we constructed a heterogeneous, multi-relational clinical knowledge graph that integrates curated ontology semantics with empirically observed clinical associations.

Empirical relationships were derived from MIMIC-IV at the level of individual hospital admissions. For each admission (hadm_id), all unique SNOMED-mapped diagnosis concepts were identified, and pairwise co-occurrence relations were established between concepts appearing within the same admission. These relations were encoded using a symmetric co_occurs_with relation type and represent clinically observed associations that are not explicitly defined within the ontology.

In parallel, the ontology-based graph component was constructed from SNOMED CT using formally curated semantic relationships. These include both hierarchical relations (e.g., is-a) and non-hierarchical relations (e.g., part-of, associated-with, morphology-of), which collectively encode expert-validated biomedical knowledge and provide the structural backbone of the ontology.

The ontology-derived relations and the empirically derived co-occurrence relations were subsequently merged into a single unified graph representation. Each relation is represented by a source concept, a destination concept, a relation type, and a provenance indicator specifying whether the relation originates from curated ontology knowledge or empirical clinical data. This design enables joint reasoning over structured medical semantics and real-world clinical associations within a unified relational learning framework suitable for Relational Graph Convolutional Network (R-GCN) training.

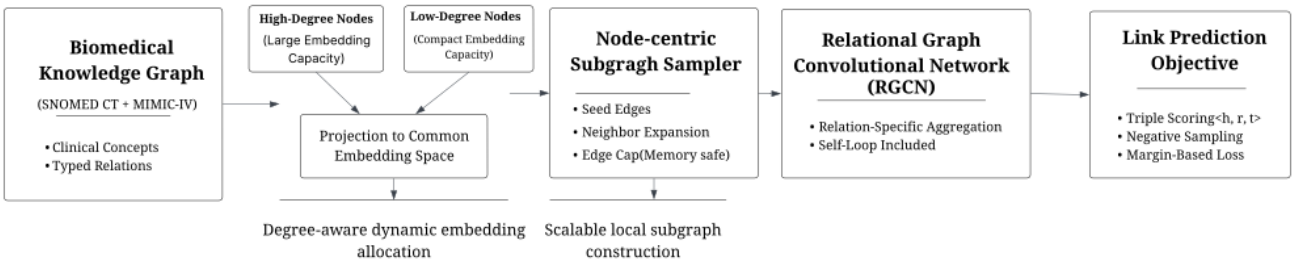

Figure 2. High-level overview of the proposed Dynamic-Capacity R-GCN framework.

To ensure semantic consistency and data quality, the integrated graph underwent systematic validation and conflict resolution. When the same concept pair was associated with multiple relation types, curated SNOMED CT relations were prioritized over empirical co-occurrence relations. Additional quality checks addressed duplicate or redundant edges, multiple diagnostic mappings converging on the same SNOMED CT concept, and missing or incomplete concept descriptions. This conservative resolution strategy preserves the integrity of the ontology while incorporating clinically meaningful empirical evidence.

Table III: Examples of Curated Ontology Relations and Empirical Clinical Co-occurrence Relations in the Integrated Knowledge Graph

sourceId	source_term	relation_term	Destination_id	destination_term	source
2776000	Delirium	Is a (attribute)	286933003	Confusional state	SNOMED_CT
2776000	Delirium	Finding site (attribute)	12738006	Brain structure	SNOMED_CT
2776000	Delirium	co_occurs_with	59621000	Essential hypertension	MIMIC_IV
2776000	Delirium	co_occurs_with	28394000	Toxic encephalopathy	MIMIC_IV
2776000	Delirium	co_occurs_with	60046008	Pleural effusion	MIMIC_IV

Table III presents representative ontology based and empirical relations centered on a shared clinical concept in the integrated knowledge graph. The SNOMED CT rows illustrate curated semantic relations associated with the concept Delirium, including hierarchical ("is-a") and attribute-based ("Finding site") links that encode its meaning within the ontology. The MIMIC-IV rows capture empirical diagnosis co-occurrence relationships involving the same concept, extracted from patient-level admissions where Delirium appears alongside other diagnoses. Integration occurs by mapping clinical diagnoses from MIMIC-IV to their corresponding SNOMED CT concepts, resulting in a single shared node that participates in both ontology-derived and empirical relations. These relations form typed edges around the shared concept node, enabling downstream relational learning models to jointly leverage structured semantic knowledge and observed clinical co-occurrence patterns.

In addition, Figure 1 provides a visual illustration of this integration process. The figure depicts how curated SNOMED CT semantic relations and empirically derived MIMIC-IV diagnosis co-occurrence relations are first constructed independently and then aligned through ICD-10 to SNOMED CT mapping via OHDSI Athena. The Delirium concept serves as an anchor node to demonstrate how ontology-derived relations (e.g., is-a, finding site) and empirical (co_occurs_with) relations converge into a unified clinical knowledge graph. This visualization clarifies the end-to-end data flow, from raw EHR diagnoses and ontology semantics to a single integrated, multi-relational graph, highlighting how both knowledge sources jointly contribute structured edges around shared clinical concepts for downstream R-GCN training.

IV. Graph Construction

To enable large-scale relational learning over the integrated SNOMED CT and MIMIC-IV datasets, the cleaned tabular data were transformed into a numerical, multi-relational graph representation suitable for training R-GCNs. This section describes the complete pipeline used to construct node and relation indices, edge tensors, adjacency structures, and supervised training splits.

A. **Dynamic Capacity node embedding allocation:** To address the extreme degree heterogeneity present in large biomedical knowledge graphs, the proposed model adopts a degree-informed embedding allocation strategy [10]. Prior to training, node degrees are computed from the full global graph and used to partition nodes into high degree and low degree groups using a fixed degree percentile threshold of 24, where nodes with degree values at or above the 80th percentile of the global degree distribution are treated as high-degree and the remaining nodes are treated as low degree. In the constructed unified clinical knowledge graph, this results in 88,675 high-degree nodes and 334,944 low degree nodes, for a total of approximately 423,619 clinical concepts. High-degree node embeddings correspond to nodes that are connected to many other nodes in the graph and participate in a large number of relationships and are initialized with 512-dimensional embeddings to capture richer connectivity patterns. In contrast, low-degree node embeddings correspond to nodes that are connected to fewer other nodes and are initialized with 128-dimensional embeddings to reduce memory usage an parameter redundancy. Rather than operating with heterogeneous embedding sizes throughout the model, embeddings from both groups are linearly projected into a shared 256-dimensional latent space prior to relational message passing, enforcing a uniform feature dimension required by R-GCN layers and

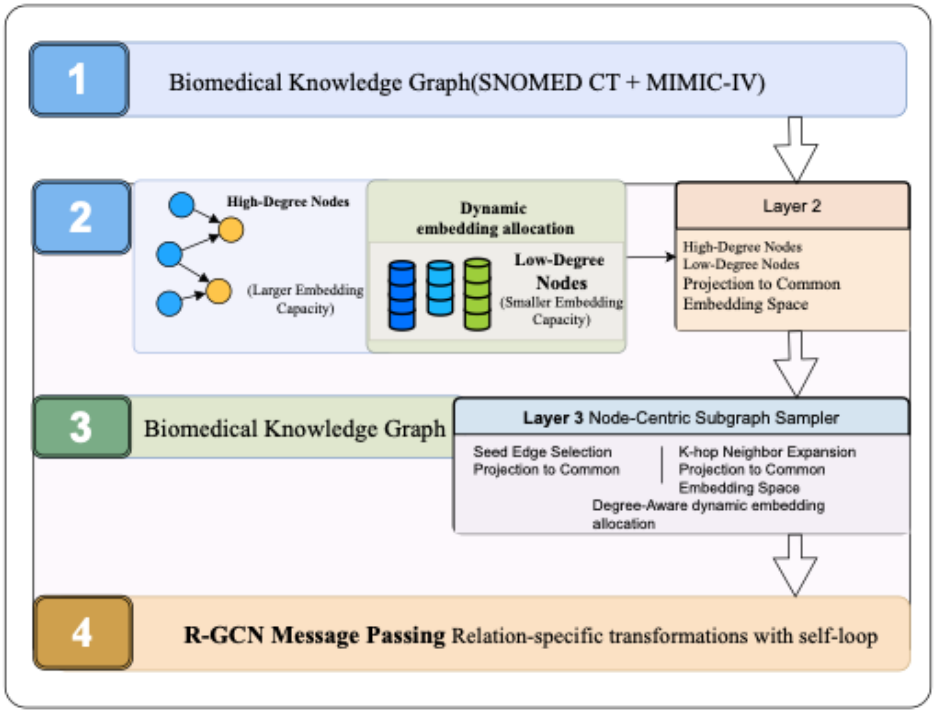

Figure 3. Training pipeline of the Dynamic-Capacity R-GCN pipeline.

enabling consistent relation-specific transformations and neighborhood aggregation across nodes.

B. **Node and relation indexing:** The unified dataset produced after data preprocessing in section 3 contains clinical concept identifiers (sourceId, destinationId) and their associate semantic relationship labels (relation_term). To construct a computable graph representation, all symbolic identifiers were first mapped to integer indices.

1. Node indexing: All unique SNOMED CT concept identifiers appearing as either a source or destination node were collected and assigned contiguous integer indices:

$$\text{conceptId} \rightarrow \text{node_index} \in [0, N),$$

where N denotes the total number of unique clinical concepts.

This mapping enables efficient storage of node representations in PyTorch tensors and supports scalable neighborhood queries during graph sampling [13].

2. Relation type indexing: All distinct semantic relationship types were enumerated, including curated SNOMED CT relations and empirical relations derived from MIMIC-IV. Each relation label was mapped to a unique integer identifier:

$$\text{relation_term} \rightarrow \text{relation_id} \in [0, R),$$

where R denotes the total number of relation types. This encoding allows R-GCN layers to maintain relation-specific transformation matrices.

C. Edge tensor construction: After node and relation indexing, each row in the cleaned dataset was converted into a directed, typed edge represented as a triplet:

(source_node_index, destination_node_index, relation_id)

The full graph was stored using two tensor structures: an edge index tensor of size $2 \times E$, where the first row contains source node indices and the second row contains destination node indices, and an edge type tensor of length E assigning a relation identifier to each edge. In addition, global statistics including the total number of unique nodes (*num_nodes*) and relation types (*num_relations*) were computed to support degree analysis and model configuration. These components form the global adjacency representation used for link prediction evaluation, negative sampling, and degree computation.

D. Train-validation-test split generation: To support supervised relational learning, the set of edges was partitioned into three mutually exclusive subsets. Approximately 80% of the edges were used for training, 10% for validation, and the remaining 10% for testing. A fixed random seed was applied during partitioning to ensure reproducibility across all experimental runs.

E. Node-Centric Adjacency Construction: Although the full graph is stored as global tensors, training does not operate on the entire graph simultaneously due to memory constraints. Instead, localized subgraphs are extracted around sampled edges during training [11]. To support this process, a node-centric adjacency structure was constructed. For each node $\{u\}$,

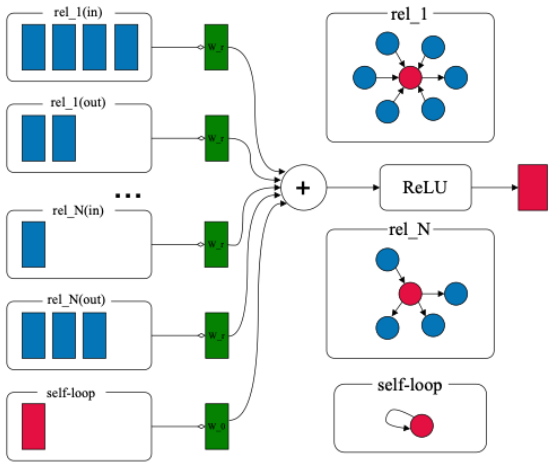

Figure 4. Relation-specific message passing in the proposed Pure R-GCN. Incoming and outgoing neighbor embeddings are transformed by relation-specific weights and aggregated with a self-loop contribution, followed by ReLU activation to update the node representation.

all outgoing edges and their associated relation types were stored as:

$$\text{node_adj[u]} = \{(r1, v1), (r2, v2), \ldots\},$$

where each tuple denotes a relation identifier and a neighboring destination node. For efficient neighborhood lookup during subgraph sampling, a node-centric adjacency representation was constructed. For each node u, the adjacency list stores a set of relation-neighbor pairs (r, v), where r denotes the relation type and v denotes the destination node. This structure enables relation-aware neighborhood expansion during R-GCN message passing. This representation enables efficient relation-aware neighborhood expansion and bounded subgraph extraction during training while enforcing strict limits on the number of nodes and edges processed per batch.

Through node and relation indexing, tensor-based adjacency construction, supervised data splits, and node-centric adjacency lists, the resulting graph representation supports scalable and memory-efficient relational learning on large biomedical knowledge graphs. It preserves distinct semantic relation types and integrates naturally with degree-aware embeddings and subgraph-based R-GCN training strategies [10], [11]. This graph representation serves as the structural foundation for the proposed Dynamic-Capacity R-GCN described in the following section.

V. Model Development

All models are trained for multi-relational link prediction on large biomedical knowledge graphs constructed from SNOMED CT and an integrated SNOMED-MIMIC dataset.

A. ComplEx baseline: ComplEx is employed as a non-graph baseline for relational learning. Each entity and relation is embedded in a complex-valued vector space, enabling the model to capture asymmetric relations commonly observed in biomedical ontologies. Given a triple, ComplEx computes the score:

$$s(h, r, t) = \Re(\langle h, r, t' \rangle), \qquad (1)$$

where $h, r, t \in C^d$, t' denotes complex conjugation, and $\Re(.)$ extracts the real part. ComplEx does not perform neighborhood aggregation or message passing and therefore serves as a fixed-embedding baseline for comparison against graph-based models.

B. Hybrid RoBERTa + R-GCN: To incorporate semantic information from concept descriptions, we evaluate a hybrid model that combines pretrained RoBERTa embeddings with relational graph convolution. Each node is initialized using a fixed textual embedding derived from its SNOMED CT description and is subsequently refined using R-GCN layers. For node i at layer l, relational message passing is defined as:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} \mathbf{W}_r^{(l)} \mathbf{h}_j^{(l)} + \mathbf{W}_0^{(l)} \mathbf{h}_i^{(l)}\right), \qquad (2)$$

where $\mathcal{N}_i^r$ denotes neighbors of node i under relation r, $\boldsymbol{W}_r^{(l)}$ are relation-specific parameters, and $\sigma(\cdot)$ is a non-linear activation. Although this hybrid approach improves semantic expressiveness, it incurs substantial memory overhead and limited scalability, motivating a structure-centric alternative.

C. R-GCN with dynamic-capacity embeddings: To enable scalable relational learning on large biomedical graphs, we propose a **R-GCN** architecture with **dynamic-capacity node embeddings**. Nodes are partitioned into high-degree and low-degree groups based on graph degree statistics. Each node embedding is defined as:

$$\mathbf{e}_i = \begin{cases} \mathbf{P}_h \mathbf{E}_h(i), & i \in \mathcal{V}_{high} \\ \mathbf{P}_l \mathbf{E}_l(i), & i \in \mathcal{V}_{low} \end{cases}, \qquad (3)$$

where $\boldsymbol{E}_h$ and $\boldsymbol{E}_l$ are learnable embeddings with different dimensionalities and $\boldsymbol{P}_h$, $\boldsymbol{P}_l$ project them into a shared latent space. Here, $E_h : V_{high} \rightarrow R^{d_h}$ and $E_l : V_{low} \rightarrow R^{d_l}$ are learnable embedding lookup tables with different dimensionalities, where $d_h > d_l$. This allows high-degree nodes, which participate in a large number of relations, to be represented with higher capacity embeddings.

Since relational graph convolution requires a uniform input dimensionality, the projection matrices

$P_h \in R^{d \times d_h}$ and $P_l \in R^{d \times d_l}$ map both embedding types into a shared latent space R^d. As a result, all nodes produce embeddings $e_i \in R^d$, enabling consistent message passing across the graph.

These projected node embeddings are then used as inputs to the stacked R-GCN layers defined in Eq. (2), allowing the model to jointly leverage high-capacity representations for structurally important nodes while maintaining computational efficiency for the remainder of the graph.

D. Subgraph-based training : Due to the scale of the graph, full-graph convolution is infeasible. Training is therefore performed on dynamically sampled **node-centric subgraphs**. Given a seed set $\mathcal{S}_0$, neighbors are expanded iteratively as:

$$\mathcal{S}_{l+1} = \bigcup_{u \in \mathcal{S}_l} \text{Sample}_K\,(\mathcal{N}(u)), \qquad (4)$$

where Sample$K(\cdot)$ selects up to K neighbors per node. Hard constraints on the number of sampled edges ensure memory-safe training while allowing exposure to diverse relational contexts.

E. Triple scoring function: All R-GCN variants use a tri-linear scoring function for link prediction:

$$s(h,r,t) = \langle \boldsymbol{h}, \boldsymbol{r}, \boldsymbol{t} \rangle = \sum_{k=1}^{d} h_k \cdot r_k \cdot t_k, \qquad (5)$$

where $\boldsymbol{h}$,$\boldsymbol{r}$, and $\boldsymbol{t}$ denote the learned embeddings of the head entity, relation, and tail entity.

F. Negative Sampling Strategy :

1) Generation of negative samples: Biomedical knowledge graphs contain only positive relations; explicit negatives are unavailable. For each observed triple: (h,r,t) a negative sample is generated by randomly corrupting the tail entity:

$$(h,r,t'), t' \sim \mathcal{U}(\mathcal{V}), \qquad (6)$$

where t' is sampled uniformly from the set of nodes and the corrupted triple does not exist in the graph. This procedure exactly matches our implementation and preserves the relation type, forcing the model to learn entity-relation compatibility.

2) Margin-Based banking loss: Training optimizes a margin-based ranking objective:

$$\mathcal{L} = \max\,(0, \gamma - s(h,r,t) + s(h,r,t')), \qquad (7)$$

where γ is a fixed margin. Randomized negative sampling introduces stochastic contrastive learning, exposing the model to diverse invalid triples across iterations, and improving generalization in incomplete biomedical graphs.

G. Model variants on the integrated graph: Two R-GCN configurations are evaluated on the integrated SNOMED-MIMIC graph, differing in neighborhood size and subgraph capacity. Results indicate that increasing subgraph capacity alters the trade-off between relational context and computational efficiency. In our experiments, smaller subgraphs achieve slightly higher-ranking performance, while larger subgraphs incur increased computational cost without consistent performance gains.

ComplEx provides a fixed-embedding baseline, the hybrid RoBERTa + R-GCN integrates textual semantics at the cost of scalability, and the R-GCN with dynamic-capacity embeddings achieves the best balance between expressiveness, scalability, and performance on large biomedical knowledge graphs.

VI. Experimental Setup

We design experiments to ensure fair comparison across baselines and proposed approaches while maintaining scalability on large biomedical knowledge graphs.

A. Task formulation

All models are evaluated on the multi-relational link prediction task. Given a triple *(h,r,t)* the objective is to learn a scoring function that assigns higher scores to valid triples than to synthetically generated invalid ones. During evaluation, link prediction is treated as a ranking problem, where the correct entity must be ranked above corrupted candidate entities generated through negative sampling.

B. Training procedure

Model training is performed using minibatch stochastic optimization. For graph-based models, each mini-batch operates on a locally sampled subgraph rather than the full

knowledge graph, enabling memory-efficient training on large-scale graphs.
All R-GCN variants employ relation-aware message passing over dynamically sampled node-centric subgraphs and are optimized using a margin-based ranking loss with randomized negative sampling. Optimization is carried out using the Adam optimizer with a fixed learning rate of 1×10^{-3}. To ensure fair comparison, all models are trained for a fixed number of epochs using consistent optimization settings.
The hyperparameters used across all models are summarized in Table IV.

C. Negative sampling

Biomedical knowledge graphs typically contain only positive relations, and the absence of a link does not guarantee biological invalidity. To train link prediction models, synthetic negative samples must be generated. In this work, we adopt a uniform tail corruption strategy: for each positive triple (h,r,t) a corresponding negative triple (h,r,t') is created by randomly replacing the tail entity t with a different entity $(h,r,t'), t' \sim \mathcal{U}(\mathcal{V})$, where V denotes the set of all entities. A 1:1 negative-to-positive ratio is used during training.

This simple corruption technique enables scalable training across multi-million-edge clinical graphs, as it is computationally lightweight and does not require domain-specific annotations. It introduces stochastic contrastive learning, encouraging the model to assign higher scores to true relations over randomly corrupted ones. Importantly, this strategy avoids imposing potentially incorrect biological assumptions by not assuming the absence of a link is truly negative.

However, uniform negative sampling has known limitations in biomedical settings. Due to ontology incompleteness, some corrupted triples may actually correspond to clinically plausible but unobserved relations, introducing false negatives. Additionally, random sampling often generates trivial negatives that can be easily distinguished from positives, especially in semantically heterogeneous domains like SNOMED CT. This can result in limited training signal, particularly for high-degree or ambiguous concepts.

Recent studies have proposed more sophisticated alternatives—such as type-constrained, degree-aware, or ontology-aware sampling—which generate harder and more informative negatives by respecting semantic constraints or graph topology. While these approaches show promise, uniform sampling remains widely adopted in large-scale biomedical benchmarks where curated negatives are unavailable. It provides a reproducible baseline and ensures consistent evaluation across models.

To balance scalability with biological plausibility, our work uses uniform sampling as a baseline, while recognizing its limitations. While this work empirically evaluates degree-aware and relation-aware alternatives, more expressive ontology-constrained or adversarial sampling strategies remain an open direction for further improving hard-negative generation.

Table IV: Hyperparameters used across Models.

Hyperparameter	Description
High degree embedding dimension	Dimensionality assigned to high-degree nodes in R-GCN, based models to capture richer relational structure (512).
Low degree embedding dimension	Reduced embedding dimensionality for low-degree nodes to improve memory efficiency without sacrificing performance (128).
Common embedding dimension	Shared latent space to which all node embeddings are projected for consistent relational message passing (256).
Hidden dimension	Size of hidden representations within R-GCN layers controlling model capacity (256).
Number of layers	Number of stacked relational graph convolution layers used for neighborhood aggregation.
Dropout rate	Dropout applied between layers to mitigate overfitting during training (0.1 for ComplEx, 0.2 for R-GCN based models).
Neighbor Sampling(K)	Maximum number of neighbors sampled per node during node-centric subgraph construction (5).
High degree Threshold	Proportion of nodes treated as high degree for dynamic embedding allocation (top 80%).
Subgraph edge cap	Upper bound on the number of edges sampled per subgraph to ensure memory-safe and scalable training (300,000).
Batch Size	Number of positive triples processed per training iteration.
Negative sampling ratio	Ratio of negative to positive triples used during training to learn discriminative link representations (1:1).
Optimizer	Optimization algorithm used to update model parameters during training (Adam).
Learning Rate	Initial step size for the optimizer controlling convergence speed (1×10^{-3}).
Training Epochs	Total number of passes over the training data (5).

To balance scalability with biological plausibility, our work uses uniform sampling as a baseline, while recognizing its limitations. While this work empirically evaluates degree-aware and relation-aware alternatives, more expressive ontology-constrained or adversarial sampling strategies remain an open direction for further improving hard-negative generation.

D. Subgraph sampling configuration : Due to the scale of the biomedical knowledge graphs, full-graph convolution is computationally infeasible. Training is therefore conducted on

node-centric subgraphs constructed dynamically for each batch. The sampling process expands local neighborhoods around seed nodes using a fixed number of neighbors per hop, subject to a hard upper bound on the total number of sampled edges to ensure memory safety.

Two configurations are evaluated on the integrated SNOMED-MIMIC graph to examine the trade-off between relational context and computational efficiency. Configuration A employs a smaller neighborhood size with a stricter edge cap, while Configuration B uses a larger neighborhood size with a higher edge cap. This comparison enables systematic analysis of scalability versus contextual coverage in large-scale relational learning.

E. Evaluation Metrics : Model performance is assessed using standard ranking-based metrics for knowledge graph completion, including Mean Reciprocal Rank (MRR), Hits@1, Hits@10, ROC-AUC, and Precision@K. These metrics jointly evaluate ranking accuracy, discriminative capability, and reliability of top-ranked predictions.

F. Implementation and Reproducibility : All models are implemented using PyTorch, which provides efficient automatic differentiation and GPU acceleration for large-scale relational graph learning. Experiments were executed on a single NVIDIA A100-SXM4 GPU with 80 GB of memory, provided by the Kaggle platform. CUDA acceleration (CUDA 12.6 with cuDNN 9.1) and mixed-precision training were enabled throughout. Hardware allocation is managed dynamically by the execution environment; therefore, specific GPU serial identifiers are not reported. GPU acceleration was required due to the scale of the integrated biomedical knowledge graph and the use of relation-specific message passing. Each experiment required approximately 12 hours of total training time for 5 epochs, corresponding to several hours per epoch. Training time varied modestly depending on subgraph capacity and negative sampling configuration; however, all models were trained under comparable computational budgets to ensure fair evaluation. Full-graph convolution over the integrated SNOMED-MIMIC knowledge graph was not memory-feasible. To address this limitation, training was performed using node-centric subgraph sampling with explicit edge limits, ensuring stable GPU memory usage while preserving sufficient local relational context.

G. Comparison of Negative Sampling Strategies

Table V: Comparison of negative sampling strategies

Negative Sampling Strategies	MRR	Hits@1	Hits@10
Uniform tail Corruption	0.8234	0.7178	0. 9919
Degree-aware Sampling	0.6440	0.4973	0.9234
Relation-aware Sampling	0.5946	0.4726	0.8212

Uniform tail corruption achieves the best overall results across all metrics validating its scalability and effectiveness for large biomedical graphs despite its simplicity. In contrast, degree-aware sampling yields moderate gains in Hits@10 but lags behind in MRR and Hits@1, suggesting that although it helps capture frequent node associations, it may struggle with fine-grained discrimination. Relation-aware sampling performs lowest across metrics, likely due to the semantic sparsity and heterogeneity of biomedical relation types, which complicates constrained sampling. These findings reinforce the practicality of uniform sampling as a reproducible baseline for large-scale biomedical graph completion, while also motivating future work on hybrid or adaptive strategies that combine the scalability of uniform sampling with the specificity of ontology-aware methods.

VII. Results and Discussion

This section presents the experimental results of all evaluated models and provides an analysis of their performance on large-scale biomedical knowledge graphs. The results highlight the effectiveness of the proposed Dynamic-Capacity R-GCN architecture and the impact of integrating empirical clinical relations.

A. Quantitative Results

The embedding-based ComplEx model achieves the lowest performance across all metrics, reflecting its inability to leverage graph structure beyond fixed embeddings. The Hybrid RoBERTa + R-GCN model improves ranking performance by incorporating textual semantics; however, gains are primarily observed in Hits@10 rather than top-ranked accuracy.

Table VI: Link prediction performance comparison across evaluated models

Model	MRR	Hits@1	Hits@10
ComplEx	0.2006	0.1575	0.2675
Hybrid RoBERTa +R-GCN	0.3799	0.1900	0.8200
R-GCN(SNOMED CT)	0.7593	0.6285	0.9970
R-GCN(Integrated)	**0.8531**	**0.7546**	**0.9970**

The Pure R-GCN trained on the SNOMED-only graph significantly outperforms both baselines, demonstrating the benefit of relation-aware message passing and degree-aware embeddings. The best performance is achieved by the R-GCN trained on the integrated SNOMED-MIMIC graph, which consistently yields the highest MRR, Hits@1, and ROC-AUC scores.

B. Comparison of R-GCN Variants

Table VII illustrates that varying the neighborhood size and subgraph edge cap significantly influences model performance. Configuration A, which uses a smaller neighborhood size of

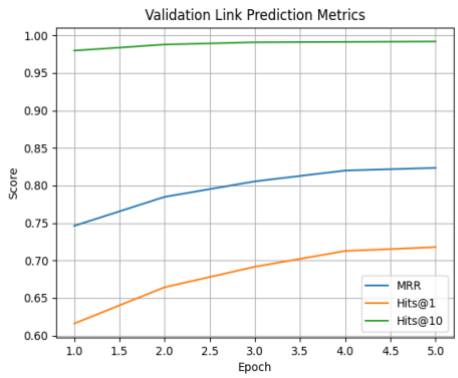

Figure 5. Validation ranking performance of the R-GCN across training epochs, measured using MRR, Hits@1, and Hits@10.

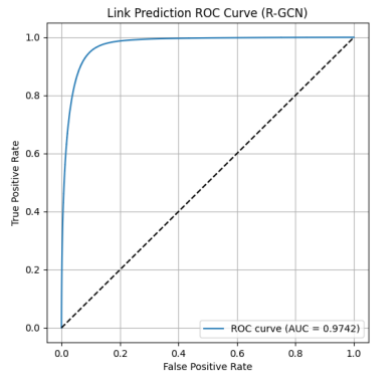

Figure 6. Receiver operating characteristics(ROC) curve for link predictions using R-GCN,

indicating strong separability between

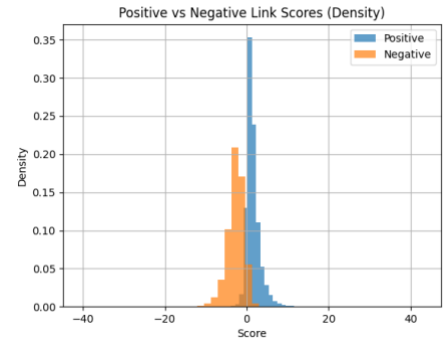

Figure 7. Distribution of learned scores for positive and negative triples, illustrating clear separation induced by

This setting captures essential local structure while minimizing noise from peripheral nodes. Configuration B, employing a neighborhood size of ten with relaxed edge constraints, underperforms suggesting that broader neighborhoods may introduce redundant or less informative relations, particularly in biomedical graphs with skewed degree distributions. Configuration C increases the neighborhood size further to fifteen and demonstrates improved performance over Configuration B, narrowing the gap with Configuration A.

Table VII: Sensitivity analysis of the proposed R-GCN under different subgraph sampling and batch configurations.

R-GCN Variants	Neighbor (K)	Edge Cap	Batch Size	MRR	Hits@1	Hits@10
Config A	5	300k	2048	0.8531	0.7546	0.9970
Config B	10	600k	4096	0.8234	0.7178	0.9919
Config C	15	600k	4096	0.8529	0.7672	0.9858

However, it incurs greater computational cost without substantial gains. These results support the use of tighter, relation-focused subgraphs in multi-relational biomedical settings. They also validate the effectiveness of node-centric subgraph sampling combined with dynamic-capacity embeddings, where high-degree nodes benefit from rich but localized context, and low-degree nodes avoid unnecessary complexity. Overall, precise tuning of sampling parameters is essential for optimizing relational generalization under memory and scalability constraints.

C. **Analysis of Ranking and Discrimination Performance** Across all Pure R-GCN variants, ROC-AUC values remain

relations from corrupted ones. Precision@K results further show that top-ranked predictions are reliable, particularly for small values of K. This is crucial in biomedical applications, where only a limited number of candidate relations are typically reviewed by domain experts.

D. Impact of Graph Integration: Integrating MIMIC-derived co-occurrence relations with the SNOMED CT ontology leads to a clear performance improvement. The empirical clinical edges introduce complementary relational signals that are not explicitly encoded in the ontology, enabling the model to learn clinically meaningful associations.

These results demonstrate that combining curated biomedical knowledge with real-world clinical data enhances relational learning and improves link prediction accuracy.

E. Discussion: The experimental results validate the design choices of the proposed Pure R-GCN architecture. Dynamic-capacity embeddings allow efficient allocation of representational power based on node connectivity, while node-centric subgraph sampling enables scalable training on large graphs.

Compared to text-enhanced models, the Pure R-GCN achieves superior performance without reliance on pretrained language models, making it more scalable and interpretable for large biomedical knowledge graphs.

F. Implications for Healthcare Informatics Practice: The proposed Dynamic-Capacity R-GCN has practical implications for healthcare informatics workflows that depend on ontology quality. First, by identifying missing or weakly defined SNOMED CT relations using empirical evidence derived from

EHR-based diagnostic co-occurrence patterns, the approach supports clinical knowledge maintenance and helps ontology curators prioritize areas where relational coverage is incomplete. Second, improving relational completeness strengthens EHR-based analytics, including cohort definition and phenotyping, where incomplete semantic links can lead to under-represented populations or inconsistent grouping of clinically related concepts. Third, more semantically consistent clinical knowledge graphs can benefit clinical decision support (CDS) pipelines that rely on relational structure to infer clinical context and trigger rule-based or learning-based recommendations, reducing failures caused by missing concept connections. Finally, integrating curated ontology structure with real-world clinical evidence supports data interoperability by improving the consistency of concept interpretation across healthcare systems and analytic pipelines.

G. Limitations

While the proposed Dynamic-Capacity R-GCN demonstrates strong performance on large-scale biomedical knowledge graphs, several limitations remain.

First, although we empirically evaluated multiple negative sampling strategies, including uniform tail corruption, degree-aware sampling, and relation-aware sampling—each approach involves trade-offs between scalability, discrimination, and biological plausibility. Uniform sampling proved most effective in our experiments, but it may still generate corrupted triples that correspond to clinically plausible but undocumented relations, potentially introducing noise during training.

Second, the current framework operates on static knowledge graphs and does not explicitly model temporal information inherent in longitudinal EHR data. As a result, clinically meaningful temporal dynamics such as disease progression, treatment sequences, or temporal comorbidities are not represented, which limits applicability for longitudinal patient modeling and time-dependent clinical decision support.

Third, the model operates in a transductive setting, assuming that all concepts and relations are observed during training. This assumption restricts generalization to unseen nodes or relation types and may limit adaptability as biomedical ontologies evolve over time.

Finally, while the inferred relations demonstrate strong performance on ranking-based evaluation metrics, the framework does not directly assess real-world clinical utility within deployed healthcare workflows. Predicted SNOMED CT links are not validated by domain experts, which limits conclusions regarding clinical plausibility, trustworthiness, and deployment readiness.

VIII. Conclusion and Future Work

In this work, we presented a scalable relational learning framework for large biomedical knowledge graphs based on a Pure Relational Graph Convolutional Network (R-GCN). The proposed approach addresses key limitations of existing methods by introducing degree-aware dynamic-capacity embeddings and node-centric subgraph sampling, enabling memory-efficient training on graphs containing hundreds of thousands of nodes and millions of relations.

Extensive experiments on SNOMED CT and an integrated SNOMED-MIMIC dataset demonstrate that the proposed Pure R-GCN consistently outperforms embedding-based baselines and text-enhanced graph models across ranking-based evaluation metrics. These results confirm that structure-centric relational learning with relation-specific message passing can effectively capture complex biomedical semantics without reliance on pretrained language models.

Moreover, integrating empirically derived clinical co-occurrence relations with curated ontology knowledge yields additional performance gains, highlighting the complementary roles of real-world clinical data and biomedical ontologies. Overall, this study demonstrates that scalable, structure-centric graph neural networks provide a robust foundation for biomedical knowledge graph completion and offer a strong basis for continued research at the intersection of clinical data integration and relational machine learning.

Future Work

While the proposed Dynamic-Capacity R-GCN addresses key limitations of existing relational learning methods for large biomedical knowledge graphs, several promising avenues remain for future research.

First, future work could explore improved negative sampling strategies that integrate structural graph properties, such as node degree and relation types, with ontology-aware constraints. Such hybrid approaches may produce harder negative samples while preserving biological plausibility and computational feasibility.

Second, extending the framework to support temporal or time-evolving knowledge graphs would enable modeling of longitudinal clinical information. Incorporating temporal dynamics could allow representation of clinically meaningful patterns such as disease progression, treatment sequences, and evolving comorbidities, which are critical for advanced clinical decision support in longitudinal patient modeling.

Third, future research should investigate inductive learning extensions capable of generalizing to unseen nodes and relation types. Such extensions would enable continuous integration of emerging clinical concepts and relations, supporting evolving biomedical ontologies without requiring full retraining.

Fourth, evaluating the real-world clinical utility of inferred relations remains an important direction. Involving domain experts in validation would be essential for assessing the clinical plausibility, trustworthiness, and deployment readiness of model predictions within operational healthcare workflows.

Finally, future work may explore multi-modal extensions that integrate textual or imaging-derived representations associated with clinical concepts. Leveraging modality-specific encoders could further enrich relational learning while preserving scalability and interpretability.

REFERENCES

[1] SNOMED International, "What is SNOMED CT," *SNOMED International*, 2025. https://www.snomed.org/what-is-snomed-ct

[2] A. Johnson *et al.*, "MIMIC-IV," *Physionet.org*, Oct. 11, 2024. https://physionet.org/content/mimiciv/3.1/

[3] X. Hao *et al.*, "Quantitatively assessing the impact of the quality of SNOMED CT subtype hierarchy on cohort queries," *Journal of the American Medical Informatics Association*, vol. 32, no. 1, Oct. 2024, doi: https://doi.org/10.1093/jamia/ocae272.

[4] A. Agrawal *et al.*, "The readiness of SNOMED problem list concepts for meaningful use of electronic health records," *Artificial Intelligence in Medicine*, vol. 58, no. 2, pp. 73–80, Jun. 2013, doi: https://doi.org/10.1016/j.artmed.2013.03.008.

[5] X. Hao, R. Abeysinghe, J. Shi, and L. Cui, "A substring replacement approach for identifying missing IS-A relations in SNOMED CT," *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 2611–2618, Dec. 2022, doi: https://doi.org/10.1109/bibm55620.2022.9995595.

[6] Rashmie Abeysinghe, F. Zheng, E. V. Bernstam, J. Shi, Olivier Bodenreider, and L. Cui, "A deep learning approach to identify missing *is-a* relations in SNOMED CT," *Journal of the American Medical Informatics Association*, vol. 30, no. 3, pp. 475–484, Dec. 2022, doi: https://doi.org/10.1093/jamia/ocac248.

[7] B. He, J. Q. Cheng, and H. Gu, "SNOMED CT ontology multi-relation classification by using knowledge embedding in neural network," *Smart Health*, vol. 36, p. 100560, Mar. 2025, doi: https://doi.org/10.1016/j.smhl.2025.100560.

[8] B. He, J. Q. Cheng, and H. Gu, "Static and Dynamic Embedding Approaches to Identify Is-A Relations in SNOMED CT," pp. 369–378, Jun. 2025, doi: https://doi.org/10.1109/ichi64645.2025.00050.

[9] A. Agrawal and K. Qazi, "Detecting modeling inconsistencies in SNOMED CT using a machine learning technique," *Methods*, vol. 179, pp. 111–118, Jul. 2020, doi: https://doi.org/10.1016/j.ymeth.2020.05.019.

[10] Z. Fan and X. Jin, "Degree-Aware Graph Neural Network Quantization," *Entropy*, vol. 25, no. 11, p. 1510, Nov. 2023, doi: https://doi.org/10.3390/e25111510.

[11] H. Zeng, H. Zhou, A. Srivastava, R. Kannan, and V. Prasanna, "GraphSAINT: Graph Sampling Based Inductive Learning Method," *arXiv:1907.04931 [cs, stat]*, Feb. 2020, Available: https://arxiv.org/abs/1907.04931

[12] W. Hu *et al.*, "Open Graph Benchmark: Datasets for Machine Learning on Graphs," *arXiv.org*, May 02, 2020. https://arxiv.org/abs/2005.00687

[13] M. Fey and J. E. Lenssen, "Fast Graph Representation Learning with PyTorch Geometric," *arXiv:1903.02428 [cs, stat]*, Apr. 2019, Available: https://arxiv.org/abs/1903.02428

[14] National Library of Medicine , "SNOMED CT," *www.nlm.nih.gov*, Sep. 18, 2019. https://www.nlm.nih.gov/healthit/snomedct/index.html

[15]Z. Zhang, X. Cao, C. Huang, L. Hu, and W. Wu, "Robust MPC-based centralized cooperative control of trajectory tracking and handling stability for distributed drive electric vehicles," Journal of the Franklin Institute, vol. 362, no. 18, p. 108217, Nov. 2025, doi: https://doi.org/10.1016/j.jfranklin.2025.108217.

[16]M. Schlichtkrull, T. Kipf, P. Bloem, R. Van Den Berg, I. Titov, and M. Welling, "Modeling Relational Data with Graph Convolutional Networks." Available: https://arxiv.org/pdf/1703.06103

[17]Y. Hu et al., "Path-based reasoning for biomedical knowledge graphs with BioPathNet," bioRxiv (Cold Spring Harbor Laboratory), Jun. 2024, doi: https://doi.org/10.1101/2024.06.17.599219.

[18]L. Cappelletti et al., "Node-degree aware edge sampling mitigates inflated classification performance in biomedical random walk-based graph representation learning," Bioinformatics Advances, vol. 4, no. 1, Jan. 2024, doi: https://doi.org/10.1093/bioadv/vbae036.

[19]A. Bernardi and L. Costabello, "Domain and Range Aware Synthetic Negatives Generation for Knowledge Graph Embedding Models," arXiv.org, 2024. https://arxiv.org/abs/2411.14858.

[20]A. Grover and J. Leskovec, "node2vec," Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Aug. 2019, doi: https://doi.org/10.1145/2939672.2939754.